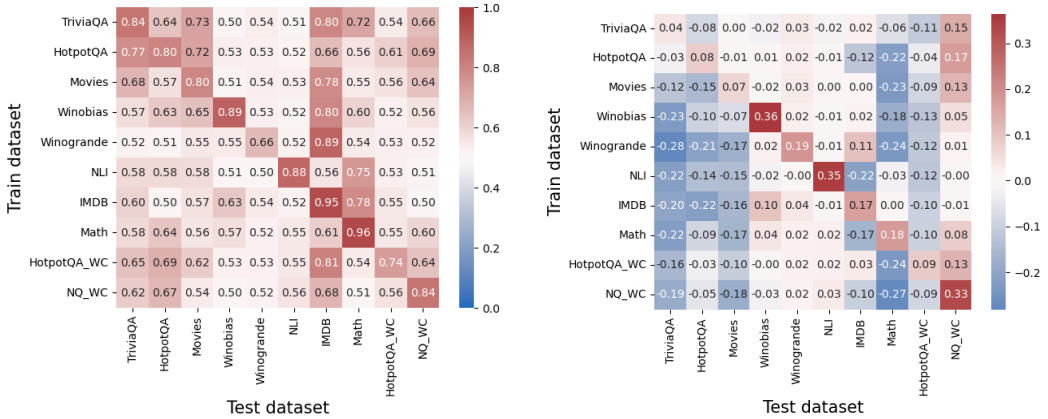
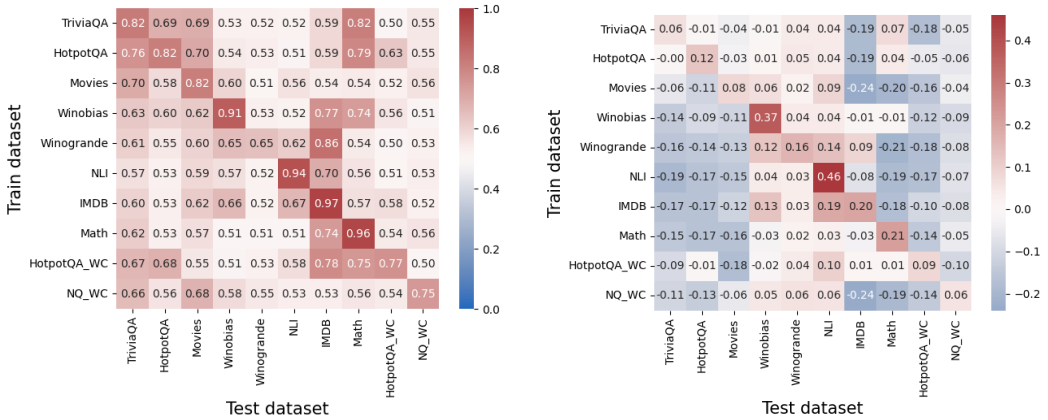


(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 7: Generalization between datasets, Mistral-7b.



(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

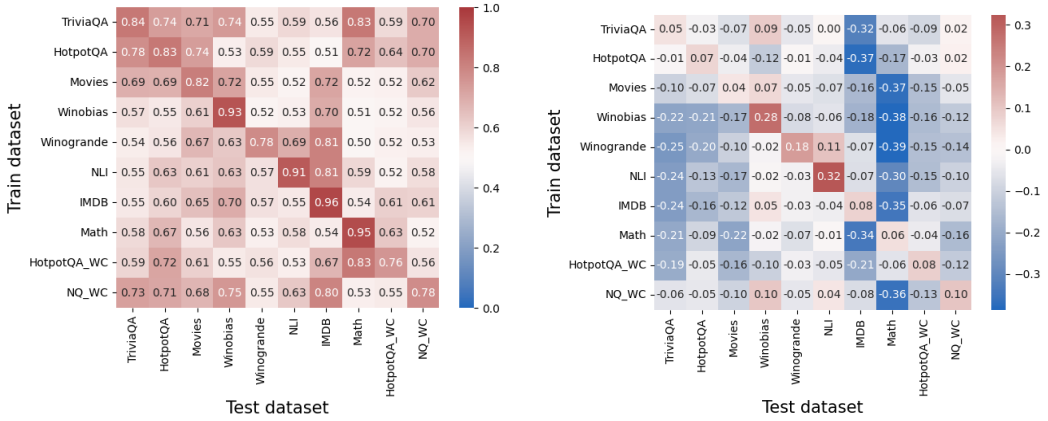
Figure 8: Generalization between datasets, Llama-3-8b.

C Full Generalization Results

Figures 7, 8 and 9 present the generalization results for the remaining models. While these results exhibit similar high-level patterns to those found in the main paper on Mistral-7b-instruct, notable differences suggest that these models may possess different mechanisms for encoding truthfulness.

D Taxonomy of Errors

Figure 10 presents, for each amount of resamples, the amount percentage of answers for which at least one generated answer was correct. The experiment was done on Mistral-7b-instruct with the TriviaQA dataset. For many answers that the greedy decoding fails to correctly provide an answer, the LLM is still able to generate the correct answer in at least one resample. The plot plateues around 30 resamples.



(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 9: Generalization between datasets, Llama-3-8b-instruct.

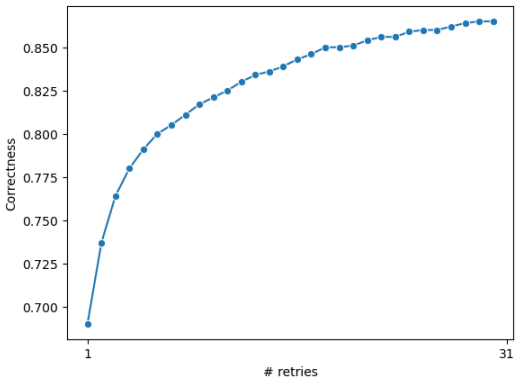


Figure 10: The percentage of answers for which at least one generated answer was correct. The first step is greedy decoding.

Table 8: AUC scores for error type classification (Winobias).

Error type	Mistral-7b	Mistral-Instr-7b	Llama3-8b	Llama3-Instr-8b
(A) Refuses to answer	-	-	-	-
(B) Consistently correct	$0.83_{\pm 0.004}$	$0.88_{\pm 0.002}$	$0.84_{\pm 0.003}$	$0.89_{\pm 0.003}$
(C) Consistently incorrect	$0.83_{\pm 0.004}$	$0.88_{\pm 0.002}$	$0.79_{\pm 0.004}$	$0.90_{\pm 0.003}$
(D) Two competing	$0.68_{\pm 0.004}$	$0.58_{\pm 0.015}$	$0.74_{\pm 0.005}$	$0.88_{\pm 0.004}$
(E) Many answers	-	-	-	-

Table 9: AUC scores for error type classification (Math). Error types are predictable from the inner model representations, indicating the encoding of fine-grained information on errors.

Error type	Mistral-7b	Mistral-Instr-7b	Llama3-8b	Llama3-Instr-8b
(A) Refuses to answer	-	-	-	-
(B) Consistently correct	$0.85_{\pm 0.017}$	$0.84_{\pm 0.007}$	$0.83_{\pm 0.020}$	$0.87_{\pm 0.006}$
(C) Consistently incorrect	$0.85_{\pm 0.026}$	$0.85_{\pm 0.003}$	$0.69_{\pm 0.032}$	$0.91_{\pm 0.007}$
(D) Two competing	-	$0.76_{\pm 0.020}$	$0.57_{\pm 0.001}$	$0.79_{\pm 0.006}$
(E) Many answers	$0.74_{\pm 0.010}$	$0.79_{\pm 0.015}$	$0.69_{\pm 0.041}$	$0.90_{\pm 0.008}$

D.1 Error Taxonomy Design Choices

The error taxonomy proposed in this paper is intentionally non-orthogonal, as some errors may simultaneously belong to multiple categories. For instance, an error might fall under both "consistently incorrect" (e.g., the same incorrect answer appears at least 15 times) and "many different answers" (e.g., the remaining answers show over 10 distinct variants).

Our taxonomy is designed to capture such nuanced cases, as restricting classification to a single category would hinder the generalizability of insights. Instead, we aim to learn general properties across different error types, providing LLM providers with actionable insights into questions exhibiting overlapping error patterns.

To support this non-orthogonal framework, our probes function as one-to-many classifiers, enabling precise error analysis and tailored solutions.

D.2 Results on Additional Datasets

Table 8 presents the results of error type classification on the Winobias dataset and Table 9 on the Math dataset.

D.3 Qualitative Examples

Tables 10 and 11 present qualitative examples of the error types in the TriviaQA and Math datasets.

E Detecting the Correct Answer Full Results

In Table 12 we present some qualitative samples from Mistral-7b-instruct, for the phenomenon we observe at error type **(C2)** Consistently incorrect but generates the correct answer at least one time. The samples in the table represent cases where the probe chose the correct answer. Table 13 compares different decoding mechanisms, including the choice via probe, on non-instruct models, and Table 14 compares on the instruct models. For all datasets and models, we observe similar conclusions to those in the main paper: significant improvement is observed for error types where the LLM shows no preference to the correct answer.